Nome _____ Cognome _____ Data ____/____/_____

Thermal physics: mock test, answers and explanations

Mark your own paper. Parts A and C: 1 mark for each correct letter. Part B: 2 marks for each question, awarded with the marking scheme printed under it. This is exactly how your teacher will mark your written tests, so read it. Write your marks in the table at the end, and list the numbers of the questions you got wrong. Total: 30 marks. Under each answer, the small print says why each wrong option is tempting: read it even when you got the question right.

Answer key, Parts A and C

1 B	**2** A	**3** C	**4** D	**5** A	**6** C
7 B	**8** D	**9** A	**10** B	**11** C	**12** D
19 B	**20** C	**21** B	**22** A	**23** D	**24** C

Part B: **13** 24 mm **14** 161 s **15** 900 J/(kg °C) **16** 264 s **17** 60 °C **18** 25 °C

A Multiple choice [12]

1. Answer B. This thermometer is wrongly calibrated, so it has two systematic errors: a zero error (it reads 2 °C in melting ice) and a scale error (96 divisions between the fixed points instead of 100). The reading is $26 - 2 = 24$ divisions above the ice point, out of 96: $24/96 \times 100 = 25$ °C.

A Corrected only the zero error (2 °C) and forgot the scale error: 96 of its degrees cover the true 100 °C. **C** Trusted the reading: a wrongly calibrated thermometer is always wrong in the same way, so it must be corrected. **D** Added the zero error instead of removing it.

2. Answer A. The ring behaves as a uniform enlargement: the hole grows by the same fraction as the outer diameter.

B The particles around the hole move further apart, so the whole outline of the ring, hole included, becomes bigger. **C** Every length on the ring, including the diameter of the hole, increases by the same fraction. **D** The edge of the hole is made of metal and moves outwards with the rest of the ring.

3. Answer C. $\Delta L \propto \Delta\theta$ for the same rod: three times the temperature rise gives three times the increase, $3 \times 2.0 = 6.0$ mm.

A The increase depends on the temperature rise: a larger rise gives a larger increase. **B** Counted only one further step of 20 °C; $60 = 3 \times 20$, so there are two further steps. **D** Squared the factor 3. The increase is proportional to the temperature rise, not to its square.

4. Answer D. During a change of state the energy goes into the potential energy of the particles, not into their kinetic energy.

A That would raise the temperature, but the temperature is constant. **B** The energy is absorbed by the sample: it is stored, not lost. **C** Particles do not change size.

5. Answer A. Evaporation is a surface process that goes on at any temperature, slowly when cold.

B That describes boiling, at the boiling point. **C** That describes boiling. **D** A puddle evaporates without any heating.

6. Answer C. Lower temperature means lower average kinetic energy: fewer and weaker collisions with the walls.

A The mass of the particles does not change. **B** The container is rigid, so the volume, and the average spacing, stays the same. **D** Particles do not change size.

7. Answer B. Same energy and same mass: P needs more energy per kilogram for each degree, so its c is larger ($c = Q/m\Delta\theta$).

A A smaller c would make P warm up more, not less. **C** With the same c both would rise by the same amount. **D** The question says that both received the same energy.

8. Answer D. In a metal, free electrons move through the solid and transfer energy by conduction much faster than vibrating particles alone.

A A larger c would make the metal warm up more slowly, not faster. **B** Expansion does not carry energy along the handle. **C** Radiation is not the main process here: the energy travels by conduction along the solid handle.

9. Answer A. Cold air sinks, warm air rises towards the unit: the whole room circulates.

B Cooling a fluid makes it denser, not less dense. **C** Air is a poor conductor; this is why it is a good insulator. **D** The main process in a fluid is convection.

10. Answer B. Infrared, visible and ultraviolet radiation travel through a vacuum.

A Space is almost empty: conduction needs particles. **C** Right process, wrong reason: there is no air in space. **D** Convection needs a fluid; there is no fluid between the Sun and the Earth.

11. Answer C. Energy is conserved: $m_1 c\,\Delta\theta_1 = m_2 c\,\Delta\theta_2$. This is the starting point for any mixture problem.

A That is true only if the two masses are equal. **B** That is true only if the two masses are equal. The masses are different. **D** Energy is conserved. The cold water has a bigger mass, so its temperature changes less, but the energy gained equals the energy lost.

12. Answer D. Newton's law of cooling: the rate is proportional to $T - T_{\text{room}}$, so the curve is steepest at the start.

A The rate depends on the difference $T - T_{\text{room}}$, which keeps falling. **B** An object cannot cool below its surroundings by losing energy to them. **C** The opposite: the rate is smallest when the difference is small.

B Short answers [12]

Marking rule, the same for every question in Part B. Mark the line only: **2 marks** for the correct final answer (the equivalent forms listed are accepted); **1 mark** for the single condition listed under 1 mark; **0 marks** for anything else. There are no marks for a method that is not on the line or in the condition listed. A question is either right, or it earns the 1 mark in its grid, or it earns nothing.

13. Answer: 24 mm

$\Delta L = \alpha L_0\,\Delta\theta = 12 \times 10^{-6} \times 80 \times 25 = 0.024\ \text{m} = 24\ \text{mm}$.

2 marks	24 mm (or 0.024 m if the unit is written)
1 mark	Correct apart from a power of ten (for example 24 000 mm or 2.4 mm).
0 marks	Anything else.

Common wrong answers. 33.6 mm: used the final temperature (35) instead of the change (25). 0.024 mm: forgot to convert metres to millimetres.

14. Answer: 161 s

$E = mc\,\Delta\theta = 1.2 \times 4200 \times 80 = 4.03 \times 10^5\ \text{J}$; $t = E/P = 4.03 \times 10^5/2500 = 161\ \text{s}$.

2 marks	161 s (accept 160 s or 1.6×10^2 s; 2.7 min if the unit is written)
1 mark	$E = 4.0 \times 10^5$ J (or 403 200 J) written on the page, whatever the final answer.
0 marks	Anything else.

Common wrong answers. 1.6×10^5 s: used the power as 2.5 W instead of 2500 W. 4.0×10^5 s: that is the energy; forgot to divide by the power.

15. Answer: 900 J/(kg °C)

$E = Pt = 100 \times 90 = 9000\ \text{J}$; $\Delta\theta = 45 - 20 = 25\ ^\circ\text{C}$; $c = E/(m\,\Delta\theta) = 9000/(0.40 \times 25) = 900\ \text{J/(kg}\,^\circ\text{C)}$.

2 marks	900 J/(kg °C)
1 mark	$E = 9000$ J written on the page, or the answer correct apart from a power of ten.
0 marks	Anything else.

Common wrong answers. 360: forgot the mass ($9000/25$). 500: used the final temperature (45) instead of the rise (25).

16. Answer: 264 s

$E = mL = 0.40 \times 3.3 \times 10^5 =$ 1.32 × 10^5 J; $\quad t = E/P = 1.32 \times 10^5/500 =$ 264 s.

2 marks	264 s (accept 260 s; 4.4 min if the unit is written)
1 mark	$E =$ 1.3 × 10^5 J (or 1.32 × 10^5 J) written on the page, whatever the final answer.
0 marks	Anything else.

Common wrong answers. 1.3 × 10^5 s: that is the energy; forgot to divide by the power. 1840 s: used the latent heat of vaporisation instead of fusion.

17. Answer: 60 °C

Energy lost = energy gained: $0.30\,(90-\theta) = 0.20\,(\theta-15)$, so $27 - 0.3\theta = 0.2\theta - 3$, $\theta = 30/0.5 =$ 60 °C.

2 marks	60 °C
1 mark	The balance equation $0.30\,(90-\theta) = 0.20\,(\theta-15)$ (or equivalent) written correctly, whatever the final answer.
0 marks	Anything else.

Common wrong answers. 52.5 °C: took the plain average of 90 and 15 (valid only for equal masses). 75 °C: added the temperatures.

18. Answer: 25 °C

$\Delta\theta = \Delta L/(\alpha L_0) = 0.60 \times 10^{-3}/(12 \times 10^{-6} \times 2.0) =$ 25 °C.

2 marks	25 °C
1 mark	Correct apart from a power of ten (for example 25 000 °C or 2.5 °C).
0 marks	Anything else.

Common wrong answers. 25 000 °C: used 0.60 mm without converting to metres. 50 °C: forgot the length of the rod (L_0).

C Alternative to practical [6]

19. Answer B. The level is halfway between 25 and 26, and the scale is read to the nearest half division: 25.5 °C. The uncertainty of an analogue reading is half the smallest printed division, ±0.5 °C. The zero error is +1.0 °C: subtract it, $25.5 - 1.0 =$ 24.5 °C. The uncertainty stays ±0.5 °C. The decimal place is there because readings are made to 0.5 °C, not because the scale has tenths.

A The zero error was not removed: the thermometer reads 1.0 °C too high at the ice point. **C** The uncertainty of an analogue reading is half the smallest division, not the whole division. **D** Added the zero error instead of subtracting it.

20. Answer C. This point is 30 counts below the line, while all the others are within 1 or 2 counts: it does not follow the trend.

A The point at 0 °C lies about 1 count from the line, like the other regular points. **B** The point at 20 °C lies about 1 count from the line. **D** The point at 40 °C lies about 1 count from the line.

21. Answer B. Read two points far apart on the line: $(610 - 210)/(50 - 0) = 8.0$ counts per degree.

A Inverted the ratio ($\Delta T/\Delta N$). **C** Divided one reading by its temperature ($610/50$). That ignores the intercept: the line does not pass through the origin. **D** That is the intercept (the value of N at 0 °C), not the gradient.

22. Answer A. From the law of cooling, $\ln(T - T_{\text{room}}) = \ln(T_0 - T_{\text{room}}) - kt$: a straight line with gradient $-k$.

B An exponential decay gives a curve. **C** $\ln t$ is not defined at $t = 0$ and does not linearise the law. **D** Still an exponential curve: only the logarithm turns it into a straight line.

23. Answer D. The line goes from 3.00 at $t = 0$ to 2.40 at $t =$ 1200 s: gradient $= -0.60/1200 =$ –5.0 × 10^{-4}/s, so $k =$ 5.0 × 10^{-4}/s.

A That is $1/k$, a time (the time constant), not k. **B** Used the change in y without dividing by the change in t. **C** The gradient is $-k$. The constant k itself is positive.

24. Answer C. The extended line meets the dashed line at about $t =$ –360 s $= -6$ min, so the potato left the bath 6 min before 10:05, at 09:59.

A That is 20 min before Start: 20 min is the length of the recording, not the time in the bath. **B** That is 12 min before Start: misread the time axis; the line meets the dashed line at about -360 s, not -720 s. **D** Sign error: a negative time means before Start, not after it.

Your marks [30]

	Your marks	Out of
Part A (questions 1–12)		12
Part B (questions 13–18)		12
Part C (questions 19–24)		6
Total		**30**

Questions I got wrong: _____

What to revise, by topic. Find the questions you got wrong.

Temperature scales and thermometers	1
Thermal expansion	2, 3, 13, 18
Changes of state and latent heat	4, 5, 16
Gases and pressure	6
Specific heat capacity and power	7, 14, 15
Mixtures: energy lost = energy gained	11, 17
Conduction, convection and radiation	8, 9, 10
Cooling and the law of cooling	12, 22, 23, 24
Laboratory skills: reading a scale, zero error, graphs	19, 20, 21